LEVEL 1 · PART I

1 Linux Fundamentals

Goal: Become comfortable operating Linux servers without depending on tutorials.

RECOMMENDED TIME	TOPICS
2-4 weeks	13

PRACTICE / CAPSTONE

Deploy Ubuntu + Nginx + Node.js API + PostgreSQL on a VPS manually, without Docker.

Topics

#	Topic	What you will learn
1	Linux Filesystem Hierarchy **p.215**	/etc /var /home /opt /usr /tmp /proc /sys and what lives where
2	Navigating and Managing Files **p.225**	ls cd pwd cp mv rm mkdir touch
3	Viewing and Searching Text **p.237**	cat less grep find, pipes and redirection
4	Text Processing with awk and sed **p.249**	awk, sed, log parsing one-liners
5	Network and Transfer Commands **p.261**	curl wget ssh scp rsync
6	File Permissions: chmod and chown **p.273**	rwx, octal, ownership, umask, setuid basics
7	Users, Groups, sudo and root **p.285**	useradd, groups, sudoers, root vs service accounts
8	Processes, PIDs and Signals **p.297**	PID, parent/child, fork/exec, SIGTERM/SIGKILL/SIGHUP, zombies
9	Daemons and systemd **p.309**	what a daemon is, systemd units, targets, writing a unit file
10	System Monitoring Commands **p.323**	ps top htop df du free uptime, load average
11	Services and Logs: systemctl and journalctl **p.335**	start/stop/enable/status, journalctl filters, log rotation
12	SSH Keys, known_hosts and authorized_keys **p.349**	keypairs, ssh-keygen, host verification, authorized_keys

#	Topic	What you will learn
13	ssh-agent and Port Forwarding **p.363**	ssh-agent, agent forwarding, local/remote/dynamic tunnels, ProxyJump, ~/.ssh/config
★	Level capstone project **p.377**	Deploy Ubuntu + Nginx + Node.js API + PostgreSQL on a VPS manually, without Docker.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

1 Linux Filesystem Hierarchy

2 Navigating and Managing Files

3 Viewing and Searching Text

4 Text Processing with awk and sed

5 Network and Transfer Commands

6 File Permissions: chmod and chown

7 Users, Groups, sudo and root

8 Processes, PIDs and Signals

9 Daemons and systemd

10 System Monitoring Commands

11 Services and Logs: systemctl and journalctl

12 SSH Keys, known_hosts and authorized_keys

13 ssh-agent and Port Forwarding

★ Capstone